## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Association of myocardial fibrosis detected by late gadolinium-enhanced MRI with clinical outcomes in patients with diabetes: a systematic review and meta-analysis |
| **AUTHORS** | Yang, Zhi; Xu, Rong; Wang, Jia-rong; Xu, Hua-yan; Fu, Hang; Xie, Ling-jun; Yang, Meng-xi; Zhang, Lu; Wen, Ling-yi; Liu, Hui; Li, Hong; Yang, Zhi-gang; Guo, Ying-kun |

## VERSION 1 – REVIEW

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| **REVIEWER** | Sultan, Fateh Ali Tipoo<br>The Aga Khan University Hospital, Medicine |
| **REVIEW RETURNED** | 05-Aug-2021 |

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| **GENERAL COMMENTS** | Very well written article |

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| **REVIEWER** | Bojer, Annemie Stege<br>Slagelse Hospital, Department of cardiology and endocrinology |
| **REVIEW RETURNED** | 12-Aug-2021 |

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| **GENERAL COMMENTS** | This study presents data from a systematic literature review and meta-analysis on LGE ability to predict serious events in patients with type 2 diabetes. This is an interesting and very relevant topic and the study is well conducted.<br><br>However, I do have some editing suggestions:<br><br>General comments<br>- I think it would strengthen the manuscript a lot if it is throughout the whole text specified if the authors mean ischemic LGE or non-ischemic LGE or a mixture of these two types. For exampel under the section Strengths and limitations of this study.<br>- Data selected on December 2019, please update<br>- Clearly state that the studied population had both type 1 and type 2 diabetes (in the abstract and introduction<br><br>Minor issues<br><br>- Line 35-37 at the end of the abstract, is this HR for MACCE or MACE? Please clarify.<br>- Please rewrite the section Strengths and Limitations of this study. You need to comment on the methods of your study here. Don't present conclusions of the findings.<br><br>Introduction<br>- Well written |

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| | - Consider committing on (could also be in the discussion section) that non-ischemic only just recently has been systematically studied. Please see: Bojer et al. Distinct non-ischemic myocardial late gadolinium enhancement lesions in patients with type 2 diabetes, Cardiovascular diabetolody. 2020;19(1):184<br><br>Methods:<br>- I have no comments for this section<br><br>Results:<br>- Consider not using the abbreviation AER<br>- Page 6 Line 23(section "prevalence of LGE and AERs), You state: However, only 3 studies reported a total of 301 patients with diabetes. I'm confused. What do you mean by this? In the section above ("study Characteristics") you state that 1121 patients had diabetes. This hole section is confusion. The AER for MACCE of 4.3% that you report here is that for all patients with diabetes both with and without LGE?? Please clearly state how many patients had LGE and how many that didn't.<br>- Page 7 First section. Please, to don't present result of the effect of having non-ischemic LGE alone, My guess is that this is because the data on the matter was to small. Please state this in the section.<br><br>Discussion<br>- In the first paragraph (page 7 line 23-30) please state that the result primarily came from ischemic LGE because of sparse data on patients with non-ischemic LGE.<br>- As your result primarily stem from patients having ischemic LGE. DO we really need the CMR to identify these patients at risk? Does this CMR scan provide us with additional information that than the knowledge's on whether these patients have ischemic heart disease or not??<br>- Page 7, line 37, You state that because the prevalence of LGE was 38% then it is important to screen these patients for fibrosis. I don't agree. As mentioned above if the CMR dos not give additional information beyond patient history of ischemic heart disease. And further, because we currently don't have any treatment option for non-ischemic LGE. We currently have no reason to screen these patients for LGE changes. You need to clarify how the CMR-LGE provided additional information beyond known history of ischemic heart disease.<br>- Again throughout the discussion, be more specific – are you addressing the issue of ischemic LGE (known or unknown ischemic lesions) or are you discussion the prevalence of non-ischemic LGE.<br>- Page 8 (line 12-14) you write: "Additionally, the myocardial fibrosis detected by LGE, especially subendocardial fibrosis, indicates more severe coronary calcium and atherosclerotic disease". I don't agree, and I don't see that you reference is backing this statement. A subendocardial LGE lesion simply indicates that this person has had an myocardial infarction that has damages the myocytes. It does not mean that this person with ischemic heart disease have more pronounced coronary calcium/atherosclerosis than a person with stable coronary disease and no history of myocardial infarction and no LGE scar.<br><br>Conclusion:<br>- The last sentence again states that this study indicates that we should use CMR in routine clinical practice to risk stratify patients |

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| | with diabetes. As you can see from my comments for the discussion section I don't think your study backs this conclusion. Firstly we still don't know if we gain any additional information from the LGE-MRI in patients with known ischemic heart disease. However in the patients without known ischemic heart disease I guess that your study suggest (if some of the patients included had unknown myocardial infarction) that a CMR screening could help identify patients in need of intensive medical therapy, but still we don't know which patients to screen. Further, in patients with non-ischemic LGE lesion – if they are indeed in increased risk of MACE/MACCE(unfortunately I don't think you have power to examine this), they would probably also need intensified medical therapy but we don't know until RCTs on the matter has been done. |

**VERSION 1 – AUTHOR RESPONSE**

Reply to the comments of Reviewer 1- Fateh Ali Tipoo Sultan's comments:
1. Very well written article.
Reply: Thank you for your high evaluation of this manuscript.

Reply to Reviewer 2- Annemie Stege Bojer's comments:
1. I think it would strengthen the manuscript a lot if it is throughout the whole text specified if the authors mean ischemic LGE or non-ischemic LGE or a mixture of these two types. For exampel under the section Strengths and limitations of this study.
Reply: Thank you for your comments. We have revised the Strength and Limitations section in our manuscript.

2. Data selected on December 2019, please update
Reply: Thank you for your suggestions. We have amended this section in our manuscript.

3. Clearly state that the studied population had both type 1 and type 2 diabetes (in the abstract and introduction.
Reply: Thank you for your advice. We have revised this in the abstract of our manuscript.

4. Line 35-37 at the end of the abstract, is this HR for MACCE or MACE? Please clarify.
Reply: Thank you for your comments. We have revised the abstract section in our manuscript.

5. Please rewrite the section Strengths and Limitations of this study. You need to comment on the methods of your study here. Don't present conclusions of the findings.
Reply: Thank you for your advice. We have revised the Strength and Limitations section in our manuscript.

6. Consider committing on (could also be in the discussion section) that non-ischemic only just recently has been systematically studied. Please see: Bojer et al. Distinct non-ischemic myocardial late gadolinium enhancement lesions in patients with type 2 diabetes, Cardiovascular diabetolody. 2020;19(1):184
Reply: Thanks for the reviewer's helpful comments and reminding us the latest study by Bojer et al. (doi: 10.1186/s12933-020-01160-y.). We have carefully read it and cited the study by Bojer et al as reference in our study.

7. Consider not using the abbreviation AER
Reply: Thanks for your reminder. The abbreviation “AER” has been changed into “annualized event rates”.

8. Page 6 Line 23(section “prevalence of LGE and AERs), You state: However, only 3 studies reported a total of 301 patients with diabetes. I’m confused. What do you mean by this? In the section above (“study Characteristics”) you state that 1121 patients had diabetes. This hole section is confusion. The AER for MACCE of 4.3% that you report here is that for all patients with diabetes both with and without LGE?? Please clearly state how many patients had LGE and how many that didn’t.
Reply: We thank the reviewer for the warm reminder. We have rewritten this sentence as follows: Additionally, 3 studies reported a total of 301 patients with diabetes, and 19.27% (n=58) patients with diabetes had LGE. These patients which with patients and LGE, with 27 events occurring over a median follow-up of 3.9 years. The annualized event rates of patients with diabetes and LGE was 11.94%.

9. Page 7 First section. Please, to don’t present result of the effect of having non-ischemic LGE alone, My guess is that this is because the data on the matter was to small. Please state this in the section.
Reply: Thanks for the reviewer’s helpful suggestion, and we state the relationship between non-ischemic fibrosis and MACCE/MACE in this section.

10. In the first paragraph (page 7 line 23-30) please state that the result primarily came from ischemic LGE because of sparse data on patients with non-ischemic LGE.
Reply: We thank the reviewer for the warm reminder. We added a sentence as follows: Moreover, the distribution of ischemic LGE seems to increase the unfavorable prognosis; however, in this study, non-ischemic LGE and MACCE/MACE in patients who with diabetes were not obtained.

11. As your result primarily stem from patients having ischemic LGE. DO we really need the CMR to identify these patients at risk? Does this CMR scan provide us with additional information that than the knowledge’s on whether these patients have ischemic heart disease or not??
Reply: We thank the reviewer for the warm reminder. We have carefully revised our manuscript for this.

12. Page 7, line 37, You state that because the prevalence of LGE was 38% then it is important to screen these patients for fibrosis. I don’t agree. As mentioned above if the CMR dos not give additional information beyond patient history of ischemic heart disease. And further, because we currently don’t have any treatment option for non-ischemic LGE. We currently have no reason to screen these patients for LGE changes. You need to clarify how the CMR-LGE provided additional information beyond known history of ischemic heart disease.
Reply: We thank the reviewer for the warm reminder. We have deleted this sentence.

13. Again throughout the discussion, be more specific – are you addressing the issue of ischemic LGE (known or unknown ischemic lesions) or are you discussion the prevalence of non-ischemic LGE.
Reply: We thank the reviewer for the warm reminder. We have rewritten and added non-ischemic LGE lesions in our discussion section.

14. Page 8 (line 12-14) you write: “Additionally, the myocardial fibrosis detected by LGE, especially subendocardial fibrosis, indicates more severe coronary calcium and atherosclerotic disease”. I don’t agree, and I don’t see that you reference is backing this statement. A subendocardial LGE lesion simply indicates that this person has had an myocardial infarction that has damages the myocytes. It does not mean that this person with ischemic heart disease have more pronounced coronary

calcium/atherosclerosis than a person with stable coronary disease and no history of myocardial infarction and no LGE scar.
Reply: We thank the reviewer for the warm reminder. We have rewritten this sentence as follows: Additionally, the myocardial fibrosis detected by LGE, especially subendocardial fibrosis, indicates patients with diabetes has had a subendocardial infarction in the past, which denotes a higher risk of MACE in the future.

15. The last sentence again states that this study indicates that we should use CMR in routine clinical practice to risk stratify patients with diabetes. As you can see from my comments for the discussion section I don't think your study backs this conclusion. Firstly we still don't know if we gain any additional information from the LGE-MRI in patients with known ischemic heart disease. However in the patients without known ischemic heart disease I guess that your study suggest (if some of the patients included had unknown myocardial infarction) that a CMR screening could help identify patients in need of intensive medical therapy, but still we don't know which patients to screen. Further, in patients with non-ischemic LGE lesion – if they are indeed in increased risk of MACE/MACCE(unfortunately I don't think you have power to examine this), they would probably also need intensified medical therapy but we don't know until RCTs on the matter has been done.
Reply: We thank the reviewer for the warm reminder. We have deleted the word of "routine".

**VERSION 2 – REVIEW**

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| **REVIEWER** | Bojer, Annemie Stege<br>Slagelse Hospital, Department of cardiology and endocrinology |
| **REVIEW RETURNED** | 08-Oct-2021 |

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| **GENERAL COMMENTS** | Dear authors.<br>Great respect for your work, however I still think you need to address a couple of major issues before this can be accepted.<br>1) If I should at all be interesting to look at LGE changes as a risk factor of MACE/MACCE it should add additional information than what is generally known about patients history. Sad in another way does the knowledge about LGE lesions add anything to the risk score compared to the knowledge that the patient has known ischemic heart disease eg. prior AMI or chronic ischemic heart disease. Therefore I suggest adding an additional adjusted analysis where this has been added as a covariable.<br>2) Throughout the discussion and conclusion It still needs to be made more clear that this is primary ischemic heart LGE lesions that you are looking at and be a sentence could be made about the fact that no studies have been made to access the prognostic impact of non-ischemic LGE lesions in patients with diabetes and that this could be a focus for future studies. |

## VERSION 2 – AUTHOR RESPONSE

Reply to Reviewer 2- Annemie Stege Bojer's comments:

1. If I should at all be interesting to look at LGE changes as a risk factor of MACE/MACCE it should add additional information than what is generally known about patients history. Sad in another way does the knowledge about LGE lesions add anything to the risk score compared to the knowledge that the patient has known ischemic heart disease eg. prior AMI or chronic ischemic heart disease. Therefore I suggest adding an additional adjusted analysis where this has been added as a covariable.

Reply: Thank you for your comments. We also have a great interest in the addictive value of identification of LGE in risk stratification compared with known ischemic heart disease. However, predictive value of known patients history or known ischemic heart for occurrence of MACE/MACCE in follow-up was not known in our identified studies so additional analysis could not be performed. Related researches are needed and we are planning to conduct a prospective study to evaluate this point.

2. Throughout the discussion and conclusion It still needs to be made more clear that this is primary ischemic heart LGE lesions that you are looking at and be a sentence could be made about the fact that no studies have been made to access the prognostic impact of non-ischemic LGE lesions in patients with diabetes and that this could be a focus for future studies.

Reply: 6 studies included in our paper reported ischemic myocardial fibrosis and 2 studies reported results both of ischemic and non-ischemic myocardial fibrosis. Date of non-ischemic myocardial fibrosis hadn't been revealed. Therefore, the pooled HR of the 8 studies should be the results of ischemic fibrosis and subsequent subgroup analysis of the 6 studies also proved this. We agree with your standpoints and made corresponding revision in discussion and conclusion.

## VERSION 3 – REVIEW

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| **REVIEWER** | Bojer, Annemie Stege<br>Slagelse Hospital, Department of cardiology and endocrinology |
| **REVIEW RETURNED** | 11-Nov-2021 |

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| **GENERAL COMMENTS** | Dear authors<br>I still have a lot of major issues with this paper. It revolves around the same issues as I have pointed out in previous revisions. Please see below.<br><br>Major revisions:<br><br>1)<br>Strengths and limitations:<br>Please remove the third statement starting with "myocardial fibrosis….." this is a conclusion and doesn't belong in this section.<br><br>Introduction:<br>2)<br>Comment for this section |

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| | "Patients with diabetes have a higher prevalence of myocardial fibrosis<br>than their nondiabetic counterparts as a result of macrovascular dysfunction, even when they are<br>asymptomatic.2-5 Moreover, the presence of myocardial fibrosis is associated with diabetic<br>cardiomyopathy.6-8 In addition, myocardial fibrosis can increase the risk of left ventricular (LV)<br>dysfunction and heart failure with preserved ejection fraction in patients with diabetes.9 10 Therefore, it<br>is important to detect myocardial fibrosis by noninvasive imaging technology for risk stratification."<br><br>I think this section is misleading, I let be explain why. The references that you use for this section are all referring to none-ischemic and mainly diffuse interstitial fibrosis. Firstly, we can't use LE to detect interstitial diffuse fibrosis in this case the T1 mapping technique and the extracellular volume would be the non-invasive method to use. Secondly. Though autopsy studies (Rubler et al. 1972(New type of cardiomyopathy associated with diabetic glomerulosclerosis. The American journal of cardiology), Regan 1975(Cardiomyopathy and regional scar in diabetes mellitus. Transactions of the Association of American Physicians), Factor et al 1980 (Clinical and morphological features of human hypertensive-diabetic cardiomyopathy. American heart journal) early on have reported on both focal and diffuse interstitial fibrosis, where the focal possibly would be detectable by LGE, this is not the scope of this review. As you are pooling both ischemic and non-ischemic findings, with most studies reporting on ischemic LGE. To the best of my knowledge, only one published study has systematically reported on non-ischemic LGE lesions (Bojer et al. 2020(Distinct non-ischemic myocardial late gadolinium enhancement lesions in patients with type 2 diabetes. Cardiovasc Diabetol)), but unfortunately, this study was cross-sectional and hence did not have information about the prognostic impact of these findings.<br>Thus you have to revise once more. You have to focus on describing why it is even relevant to look at LGE changes as a risk biomarker instead of just using a diagnosis of known ischemic heart disease in these patients. Perhaps focus on the higher prevalence of silent ischemia in DM patients. You can also use the paragraph above and simply make it clear that focal non-ischemic fibrosis possibly also are present in DM patients and possible could be a risk marker but that there currently is very little research being done on this matter.<br><br>3)<br>Paragraph 2 in the introduction:<br>Please make it clear that LGE CMR ONLY can detect FOCAL ischemic and non-ischemic Fibrosis, for non-invasive assessment of diffuse fibrosis T1-mapping is needed.<br><br>4)<br>When you refer to reference 2, please make it clear that this is unrecognized ischemic fibrosis that was assessed.<br><br>5)<br>At results section:<br>You state that references 17 reports on non-ischemic LGE but as far as I can see what they report on are unrecognized ischemic |

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| | scars. Hence, silent myocardial infarctions. In the conclusion, they write quote "… high prevalence of myocardial scar consistent with MI…."<br><br>6)<br>Discussion:<br>Please make it clear that we still don't know if performing LGE assessment brings any additional information beyond classic risk stratifying with for example the ADVANCE risk score together with known CVD.<br><br>7)<br>Conclusion<br>1) Please revise again. Please make it clear that future research is needed to confirm it LGE adds to a risk stratifying of DM patients that goes beyond classic risk stratifying with for example the ADVANCE risk score together with known CVD.<br><br>Minor corrections<br>Abstract:<br>In the design section: Please remove "accordance with" as this has been written twice. |

**VERSION 3 – AUTHOR RESPONSE**

Reply to Reviewer 2- Annemie Stege Bojer's comments:
1. Strengths and limitations:
Please remove the third statement starting with "myocardial fibrosis….." this is a conclusion and doesn't belong in this section..
Reply: Thank you for your suggestions. We have deleted this sentence.

2. Introduction:
Comment for this section
"Patients with diabetes have a higher prevalence of myocardial fibrosis
than their nondiabetic counterparts as a result of macrovascular dysfunction, even when they are asymptomatic.2-5 Moreover, the presence of myocardial fibrosis is associated with diabetic cardiomyopathy.6-8 In addition, myocardial fibrosis can increase the risk of left ventricular (LV) dysfunction and heart failure with preserved ejection fraction in patients with diabetes.9 10 Therefore, it is important to detect myocardial fibrosis by noninvasive imaging technology for risk stratification."

I think this section is misleading, I let be explain why. The references that you use for this section are all referring to none-ischemic and mainly diffuse interstitial fibrosis. Firstly, we can't use LE to detect interstitial diffuse fibrosis in this case the T1 mapping technique and the extracellular volume would be the non-invasive method to use. Secondly. Though autopsy studies (Rubler et al. 1972(New type of cardiomyopathy associated with diabetic glomerulosclerosis. The American journal of cardiology), Regan 1975(Cardiomyopathy and regional scar in diabetes mellitus. Transactions of the Association of American Physicians), Factor et al 1980 (Clinical and morphological features of human hypertensive-diabetic cardiomyopathy. American heart journal) early on have reported on both focal and diffuse interstitial fibrosis, where the focal possibly would be detectable by LGE, this is not the scope of this review. As you are pooling both ischemic and non-ischemic findings, with most studies reporting on ischemic LGE. To the best of my knowledge, only one published study has systematically

reported on non-ischemic LGE lesions (Bojer et al. 2020(Distinct non-ischemic myocardial late gadolinium enhancement lesions in patients with type 2 diabetes. Cardiovasc Diabetol)), but unfortunately, this study was cross-sectional and hence did not have information about the prognostic impact of these findings.

Thus you have to revise once more. You have to focus on describing why it is even relevant to look at LGE changes as a risk biomarker instead of just using a diagnosis of known ischemic heart disease in these patients. Perhaps focus on the higher prevalence of silent ischemia in DM patients. You can also use the paragraph above and simply make it clear that focal non-ischemic fibrosis possibly also are present in DM patients and possible could be a risk marker but that there currently is very little research being done on this matter..
Reply: Thank you for your suggestions. We have revised this section, and rewritten this paragraph as follows: Diabetes is becoming a global health care problem, and it is estimated that there will be 693 million individuals with diabetes by 2045. Patients with diabetes have a higher prevalence of ischemic myocardial fibrosis and non-ischemic myocardial fibrosis than their nondiabetic counterparts, and the mechanism has been confirmed extensively. The phenotype of unrecognized ischemic myocardial fibrosis in patients with diabetes was well studied and was associated with 4-8 folds increase in the risk of major adverse cardiac events (MACEs). However, even without myocardial ischaemia, hyperglycaemia, oxidative stress, and inflammation may lead to diffuse interstitial and non-ischemic myocardial fibrosis in patients with diabetes.6-8 In addition, diffuse interstitial myocardial fibrosis can increase the risk of non-ischemic myocardial fibrosis, and was associated with increased risk of left ventricular (LV) dysfunction in patients with diabetes. However, Non-ischemic myocardial fibrosis, may be a biomarker for risk stratification, has not been systematically characterized.

3.
Paragraph 2 in the introduction:
Please make it clear that LGE CMR ONLY can detect FOCAL ischemic and non-ischemic Fibrosis, for non-invasive assessment of diffuse fibrosis T1-mapping is needed.
Reply: We appreciated your warm reminding sincerely. We have carefully revised our manuscript for this.

4. When you refer to reference 2, please make it clear that this is unrecognized ischemic fibrosis that was assessed.
Reply: Thanks for the reviewer's helpful suggestion, and we have carefully modified this.

5.At results section:
You state that references 17 reports on non-ischemic LGE but as far as I can see what they report on are unrecognized ischemic scars. Hence, silent myocardial infarctions. In the conclusion, they write quote "… high prevalence of myocardial scar consistent with MI….".
Reply: We thank the reviewer for the warm reminder. For references 17, in the study by Kwong et al. (2008) diabetic patients with clinical evidence of MI were defined as the control group and those without clinical evidence of MI were defined as the study group, the survival analysis was actually performed in the study group which was based on the comparison between LGE positive and LGE negative patients. Furthermore, we carefully read this study, and considered the myocardial scars detected by LGE-MRI in the study group were ischemic myocardial fibrosis. Therefore, we have revised the results Section of the article.

6. Discussion:
Please make it clear that we still don't know if performing LGE assessment brings any additional information beyond classic risk stratifying with for example the ADVANCE risk score together with known CVD.
Reply: Thank you for your comments. We have amended this section in our manuscript.

7. Conclusion
1) Please revise again. Please make it clear that future research is needed to confirm it LGE adds to a risk stratifying of DM patients that goes beyond classic risk stratifying with for example the ADVANCE risk score together with known CVD..
Reply: Thank you for your comments. We have been modified this section as follows: In patients with diabetes, the presence of myocardial fibrosis detected by LGE-MRI, especially ischaemic lesions, was markedly associated with an important and increased risk of MACCEs/MACEs. This meta-analysis highlights the role of LGE-MRI in helping predict MACCEs/MACEs in complicated diabetic patients, especially those with cardiac complications and a high risk for myocardial fibrosis. Although we reported that ischaemic myocardial fibrosis detected by LGE-MRI is a strong risk marker for improving risk stratification in patients with diabetes, whether LGE-MRI provides incremental prognostic information to MACCEs/MACEs over risk stratification by conventional cardiovascular risk factors requires future study.

8. Conclusion
In the design section: Please remove ”accordance with” as this has been written twice.
Reply: Thank you for your suggestions. We have deleted “accordance with” in our manuscript**.**

**VERSION 4 – REVIEW**

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| **REVIEWER** | Bojer, Annemie Stege<br>Slagelse Hospital, Department of cardiology and endocrinology |
| **REVIEW RETURNED** | 17-Dec-2021 |

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| **GENERAL COMMENTS** | I will now recommend your manuscript for acceptance.<br>Most importantly you have now addressed that we still need further studies to tell us I in fact this information adds anything to the risk assessment of the patients. |